Mathematical Notation Reference

Measure Theory, Set Theory, Probability & Statistics

Personal Reference

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1 Introduction

This document serves as a reference for mathematical notations used in Measure Theory, Set Theory, Probability, and Statistics.

2 Set Theory

Notation	Explanation	Purpose
$x \in A$	Element x belongs to set A.	Basic set membership.
$A \subseteq B$	Set A is a subset of set B.	Indicates inclusion.
$A \cap B$	Intersection of sets A and B (elements common to both).	Logical AND operation on sets.
$A \cup B$	Union of sets A and B (elements in either A or B).	Logical OR operation on sets.
$\emptyset$	The empty set containing no elements.	Identity element for union.
2^A or $\mathcal{P}(A)$	Power set of A (set of all subsets of A).	Defining sigma-algebras.
$\forall$	Universal Quantifier (For all)	Defining properties that must hold for every element (e.g., continuity).
$\exists$	Existential Quantifier (There exists)	Asserting the existence of a specific object (e.g., existence of a limit).
$f^{-1}(B)$	Inverse Image	The core tool for defining Random Variables (Mapping values back to events).

3 Measure Theory

Notation	Explanation	Purpose
μ, ν	General Measures	Generalizing the concept of "size" (volume, mass, or count).

Notation	Explanation	Purpose
λ	Lebesgue Measure	The standard way to measure length/area/volume on the real line $\mathbb{R}$.
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra	The standard collection of "well-behaved" sets on $\mathbb{R}$ used in statistics.
$\nu \ll \mu$	Absolute Continuity	A condition ensuring that if μ sees nothing (zero), ν also sees nothing.
$\frac{d\mu}{dP}$	Radon-Nikodym Derivative	Formally defining the Probability Density Function (PDF) as a ratio of measures.
$\int_A f \, d\mu$	Lebesgue Integral	Calculating the "average" or "total" over a set A using measure μ.

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4 Probability & Statistics

Notation	Explanation	Purpose
Ω	Sample Space	The set of all possible "raw" outcomes in an experiment.
$\mathcal{F}$	σ-algebra	The collection of "measurable" events or the "information structure" available.
P	Probability Measure	A function that assigns a value in $[0, 1]$ to every event in $\mathcal{F}$.
$\nVdash_A$	Indicator Function	Converting a set A into a function (1 if inside, 0 if outside) for calculations.
$X_n \xrightarrow{a.s.} X$	Almost Sure Convergence	Proving the Strong Law of Large Numbers (the most rigorous convergence).
$X_n \xrightarrow{P} X$	Convergence in Probability	The standard convergence for consistent estimators in large samples.
$X_n \xrightarrow{d} X$	Convergence in Distribution	The foundation of the Central Limit Theorem (limit behavior of distributions).

Notation	**Explanation**	**Purpose**
θ	Parameter	The "hidden truth" or the target value in a statistical model.
$\hat{\theta}_n$	Estimator	The rule or formula used to guess θ based on n data points.
$\mathbb{E}[X\vert\mathcal{G}]$	Conditional Expectation	Predicting X given a specific sub-information (sub-σ-algebra) $\mathcal{G}$.